Segmentation Analytics Workshop

Factor analysis, clustering, classification, and embedding-based segmentation · Week 3

Marketing Principle #1: All customers differ

Yixing Chen

Agenda

01 **Review**
Customer heterogeneity, STP, and the segmentation analytics sequence.

02 **Factor analysis workshop**
Turn overlapping survey items into interpretable need dimensions.

03 **Cluster analysis workshop**
Discover customer groups and evaluate a candidate solution.

04 **Discriminant and classification workshop**
Assign new customers and assess classification accuracy.

05 **AI Frontier 1**
Use text embeddings to incorporate open-ended customer language.

06 **Integration**
Compare structured, text-based, and fused segmentation.

Review: All Customers Differ

Customers differ in needs, constraints, identity, prior experience, and response to marketing activity.

- Observed purchases may hide differences when the market offers few meaningful alternatives.
- A difference matters when it implies a different offer, message, price, channel, or service model.
- Specialists can enter by serving a dissatisfied subgroup better than a broad provider.
- The goal is not to maximize the number of segments. The goal is to identify groups that support better decisions.

Segmentation manages customer heterogeneity. It does not eliminate it.

Review: Segmenting, Targeting, and Positioning

01

Segment

Group customers with similar needs and verify that needs differ across groups.

02

Target

Compare market attractiveness with competitive strength and resource requirements.

03

Position

Choose the benefit and evidence that create a credible relative advantage.

Needs define

Use needs and benefits to create the groups.

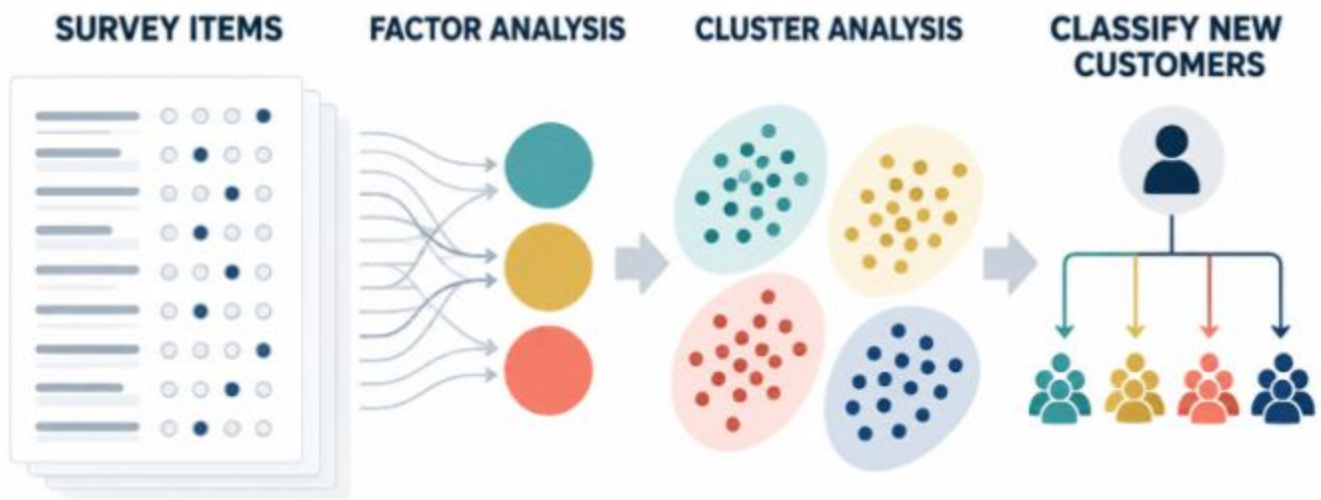
Descriptors locate

Use observable characteristics to find and reach members.

Evidence validates

Test whether groups differ on value, response, and feasible action.

Review: Segmentation Analytics Sequence



The methods answer different questions: What dimensions organize customer needs? Which customers look similar? How can the firm assign new customers to the discovered groups?

Agenda

01 Review
Concepts and analytical sequence

02 **Factor analysis**
Problem, SPSS workflow, and interpretation

03 Cluster analysis
Problem, SPSS workflow, and evaluation

04 Classification
Problem, SPSS workflow, and cross-validation

05 Embedding-based segmentation
Customer language and fused representations

06 AI Frontier summary
Opportunities, risks, and decisions

Factor Analysis: UrbanCart's Survey Problem

- UrbanCart wants to redesign its loyalty program around customer needs.
- The survey contains 12 preference questions, many of which are strongly related.
- Using all 12 items directly can overweight needs measured by several similar questions.
- Managers also need a concise language for discussing what customers value.

Workshop question

Can we represent 12 survey items with a small number of interpretable latent dimensions?

Data: 320 fictional loyalty members, 12 seven-point items, customer behavior, and open-ended comments.

Factor Analysis: SPSS Workflow

1. Import the CSV, confirm numeric types, inspect missing values, and review item distributions.
2. Inspect the correlation matrix. Factor analysis needs meaningful shared variation.
3. Check KMO and Bartlett's test before interpreting the solution.
4. Extract four factors with principal-axis factoring and use Varimax rotation for classroom interpretation.
5. Read the rotated loading matrix. Name each factor from the items that load together.
6. Save regression factor scores for every customer and inspect their distributions.

Dataset · SPSS syntax

Factor Analysis: What We Should Learn

Convenience

Speed, easy ordering, accurate availability, and simple fulfillment.

Quality

Durability, trustworthy products, knowledgeable service, and problem resolution.

Value

Low prices, savings, promotions, and budget efficiency.

Innovation

New products, digital features, early access, and personalized discovery.

- A factor is a statistical summary. Its label remains a managerial interpretation.
- Cross-loadings may indicate an ambiguous question rather than a new factor.
- Saved factor scores become comparable inputs for the clustering workshop.

Cluster Analysis: UrbanCart's Grouping Problem

- Factor scores describe each customer's relative preference for convenience, quality, value, and innovation.
- UrbanCart needs groups that differ enough to justify different loyalty benefits.
- The analyst must choose the number of clusters and check whether the solution is stable and interpretable.
- Cluster membership alone does not determine which group should be targeted.

Workshop question

Which grouping best balances statistical separation, simplicity, and managerial usefulness?

We will compare hierarchical evidence with a four-cluster k-means solution.

Cluster Analysis: SPSS Workflow

1. Open the `.sav` file created by the factor-analysis syntax and verify the four saved factor scores.
2. Run hierarchical clustering with Ward's method and squared Euclidean distance.
3. Inspect the dendrogram and agglomeration schedule for meaningful jumps in merger distance.
4. Compare plausible values of k. Avoid choosing a solution from one statistical signal alone.
5. Run four-cluster k-means and save cluster membership and distance to the assigned center.
6. Profile clusters with spend, frequency, digital share, discount use, tenure, and satisfaction.

Dataset · SPSS syntax

Cluster Analysis: What Makes a Useful Solution?

Statistical checks

- Within-cluster similarity and between-cluster separation
- Reasonable group sizes and no cluster driven only by extreme values
- Stability across starting values and small changes in the sample
- Consistent patterns in variables not used to create the groups

Managerial checks

- A distinct need pattern that the firm can address
- A practical method to identify and reach members
- Different economics or response to marketing action
- Names that summarize evidence rather than stereotypes

A cluster becomes a market segment only after it supports a different and valuable action.

Discriminant and Classification Analyses: Assignment Problem

- The research survey identifies segments, but most future customers will not complete the full 12-item instrument.
- UrbanCart already observes age, income, spend, frequency, digital share, discount share, tenure, and satisfaction.
- Management wants a practical assignment rule for CRM and campaign execution.

Workshop question

Can observable customer characteristics classify new customers with acceptable accuracy?

Misclassification costs matter. A wrong assignment can produce an irrelevant benefit or an uneconomic offer.

Discriminant and Classification Analyses: SPSS Workflow

1. Use the known synthetic segment code as the group variable.
2. Use observable descriptors and behavior as predictors. Keep the original need items out of the assignment rule.
3. Inspect group means, Box's M, standardized coefficients, and structure correlations.
4. Review canonical functions to understand which characteristics separate the groups.
5. Compare original and leave-one-out cross-validated classifications.
6. Read the confusion matrix by segment. Overall accuracy can hide a weak result for one important group.

Dataset · SPSS syntax

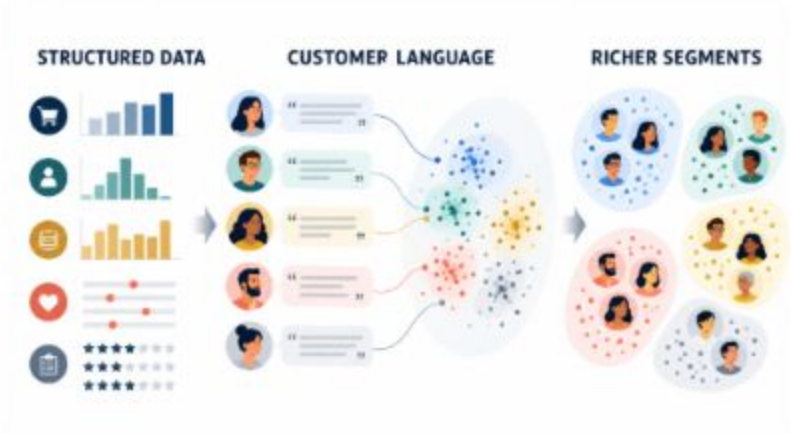
AI FRONTIER 1

Embedding-Based Segmentation

Embeddings convert open-ended customer language into numeric representations that preserve semantic similarity.

The opportunity is to recover needs that predefined survey questions did not anticipate.

Strengthening the UrbanCart Cluster Analysis with Embeddings



Structured baseline

Cluster standardized survey and behavioral variables.

Text representation

Encode each open-ended comment with a sentence-transformer model.

Fused solution

Combine normalized structured features and embeddings, then compare separation, stability, and business meaning.

Python workshop code · Sentence Transformers documentation

AI Frontier 1: Takeaways

- Embeddings add customer meaning that fixed survey questions may omit, but the dimensions are not directly interpretable.
- Structured variables and customer language provide complementary evidence. A fused representation can strengthen or distort the segmentation.
- Model choice, text length, language, and preprocessing change the geometry of the embedding space.
- Representative comments help name groups, but labels require human review and should remain provisional.
- Privacy review must remove personal and sensitive information before customer text enters the workflow.
- The final test remains managerial: do the groups differ in value, response, reachability, and feasible action?

Workshop data are synthetic. Embedding implementation follows the Sentence Transformers quickstart and scikit-learn clustering workflow.